INCH-POUND

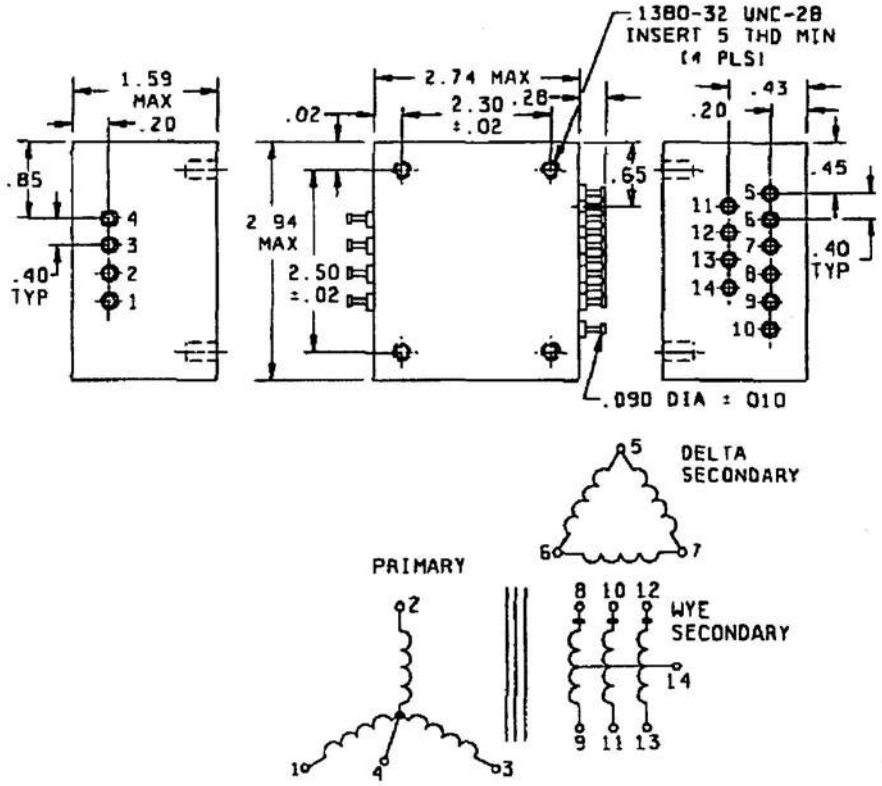
MIL-PRF-27/364A
21 June 2010
SUPERSEDING
MIL-PRF-27/364
4 December 1986

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 107 VOLTAMPERES, 3-PHASE

This specification is approved for use by all Departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the latest issue of MIL-PRF-27.



INCHES	MM
.010	0.25
.02	0.5
.090	2.29
.1380	3.505
.20	5.1
.28	7.1
.40	10.2
.43	10.9
.45	11.4
.65	16.5
.85	21.6
1.59	40.4
2.30	58.4
2.50	63.5
2.74	69.6
2.94	74.7

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the sides of the case. Terminal identification shall be as shown on the figure.
4. Electrical values shall be marked as specified in table I.
5. For series output connect terminals 4 and 5 together. For parallel output connect pins 3 to 4 and 5 to 6.

FIGURE 1. Dimensions and configuration.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Primary voltage at 400 ±20 Hz.

Wye connected:

(1-4): 115 V ac.
(2-4): 115 V ac.
(3-4): 115 V ac.

Secondary voltage: See table I.

Secondary current: See table I.

Working voltage (peak): 535 volts peak maximum.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Epoxy fiberglass.

Terminals: Solderable turret type.

Height: 0.28 ±.040 inch (7.11 ±1.02 mm).

Diameter: .090 ±.010 inch (2.29 ±0.25 mm).

Weight: 1.6 pounds, maximum.

Operating temperature range: -55°C to +130°C.

Altitude: 70,000 feet, maximum.

Terminal strength: MIL-STD-202, method 211, test condition A, 5 pounds.

Dielectric withstanding voltage:

At atmospheric pressure (each winding): 1,500 V rms.

At barometric pressure (each winding): 670 V rms.

Electrical characteristics: See table II.

Rated load: With rated voltage across the primary at 400 Hz, the secondary voltages and currents shall be as specified in table I.

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Regulation: Shall not exceed 10 percent when computed as follows:

$$\frac{\text{Voltage (no load) - Voltage (rated load)}}{\text{Voltage (rated load)}} \times 100$$

TABLE I. Electrical ratings.

Dash no.	Delta secondary (L-L)			Wye secondary (L-C)					
				Terminals 8, 10, 12 to center tap (14)			Terminals 9, 11, 13 to center tap (14)		
	Voltage (V rms)	Load current (A rms)	DC resistance (ohms)	Voltage (V rms)	Load current (A rms)	DC resistance (ohms)	Voltage (V rms)	Load current (A rms)	DC resistance (ohms)
01	22 ±2.0	1.0	.38 ±25%	19 ±1.0	.578	.50 ±25%	19 ±1.0	.144	.83 ±25%

DC resistance:

Primary:

(1-4): 3.25 ohms ±25 percent.
(2-4): 3.25 ohms ±25 percent.
(3-4): 3.25 ohms ±25 percent.

Secondary: See table I.

Temperature rise: With rated voltage across the primary at 400 Hz, secondary voltages and currents as specified in Table I, and a maximum ambient temperature of +80˚C, the temperature rise shall not exceed +50˚C.

Shock (specified pulse): MIL-STD-202, method 213.

Marking location: See figure 1.

Part or Identifying Number (PIN): M27/364-01.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents.

MIL-PRF-27
MIL-STD-202

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Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities
Army – CR4, MI
Navy – AS, MC, OS
Air Force – 19, 99

Preparing activity:
DLA - CC

(Project 5950-2009-059)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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